

Distributions

COMPLETE LEARNING GUIDE

What You'll Learn

A probability distribution describes how likely each possible outcome of a random variable is — and recognizing which distribution a real-world quantity roughly follows is one of the most useful pattern-matching skills in statistics, because it immediately tells you what questions you can answer about it and which formulas/tests apply. This submodule covers the three distributions you'll encounter constantly: the normal distribution (the familiar symmetric 'bell curve' that models things like measurement error and heights), the binomial distribution (which models the count of successes in a fixed number of independent yes/no trials, like coin flips or conversion events), and the Poisson distribution (which models the count of events happening in a fixed interval of time or space, like customer arrivals per hour). Each section includes the distribution's parameters, what it's used for, and a working `scipy.stats` code example.

1. What a Probability Distribution Is

A probability distribution assigns a likelihood to every possible value (or range of values) a random variable can take. For discrete variables (things you count, like number of defects), this is a probability mass function; for continuous variables (things you measure, like height or time), it's a probability density function, where probability corresponds to the area under the curve over a range rather than the height at a single point. Every distribution is fully described by a small number of parameters — knowing the distribution's family and its parameters tells you its mean, its spread, and lets you compute the probability of any outcome or range of outcomes without needing the raw data at all.

2. The Normal Distribution: The Bell Curve

The normal (Gaussian) distribution is symmetric and bell-shaped, fully described by just two parameters: the mean (μ) and the standard deviation (σ). It shows up constantly because of the Central Limit Theorem — the sum or average of many independent random quantities tends toward a normal distribution regardless of the shape of the original data, which is why measurement errors, aggregated survey results, and many natural quantities are approximately normal. The empirical '68-95-99.7 rule' is the key intuition to memorize: about 68% of values fall within 1 standard deviation of the mean, about 95% within 2, and about 99.7% within 3 — which is also the basis for z-scores, a standardized way to express 'how many standard deviations from the mean' any given value is.

PYTHON

```
from scipy import stats

# Heights: mean 170cm, std 7cm
dist = stats.norm(loc=170, scale=7)

print(dist.pdf(170))          # density at the peak
print(dist.cdf(177))          # P(height <= 177) = P(within 1 std above mean) "H
0.841
print(dist.cdf(177) - dist.cdf(163)) # P(163 <= height <= 177) "H 0.683 (the '68'
in 68-95-99.7)
```

3. The Binomial Distribution: Counting Successes in Fixed Trials

The binomial distribution models the number of successes in a fixed number (n) of independent trials, each with the same probability of success (p) — the canonical example is 'how many heads in 10 coin flips,' but the same shape applies to 'how many of 500 emails get opened' (if each email has the same open probability) or 'how many of 20 manufactured parts are defective.' Its two parameters, n and p , fully determine its shape, mean ($n \times p$), and variance ($n \times p \times (1-p)$). It's discrete — you can only get whole-number outcomes (0, 1, 2, ... successes), never a fraction.

PYTHON

```
from scipy import stats

# 10 coin flips, fair coin
n, p = 10, 0.5
dist = stats.binom(n, p)

print(dist.pmf(5))          # P(exactly 5 heads) "H 0.246
print(dist.cdf(5))          # P(5 or fewer heads) "H 0.623
print(dist.mean())          # expected number of heads =  $n \times p = 5.0$ 
```

4. The Poisson Distribution: Counting Events Over an Interval

The Poisson distribution models the number of times an event occurs in a fixed interval of time or space, assuming events happen independently and at a constant average rate — classic examples are 'number of customer support tickets per hour,' 'number of typos per page,' or 'number of server errors per minute.' It has a single parameter, λ , — both its mean and its variance — a distinctive property no other common distribution shares. The Poisson distribution is closely related to the binomial: it's what a binomial distribution converges to when n is very large and p is very small but $n \times p$ stays constant (many possible 'trials,' each individually very unlikely to produce the event, but observed over the whole interval).

PYTHON

```
from scipy import stats

# Support desk averages 4 tickets per hour
lam = 4
dist = stats.poisson(lam)

print(dist.pmf(6))    # P(exactly 6 tickets in an hour) "H 0.104
print(dist.cdf(2))    # P(2 or fewer tickets in an hour) "H 0.238
print(1 - dist.cdf(7)) # P(more than 7 tickets – a busy-hour event) "H 0.051
```

5. Choosing the Right Distribution for Your Data

Matching data to a distribution starts with the question 'what kind of thing am I measuring?' — a continuous, roughly symmetric measurement (heights, test scores, measurement error) suggests normal; a count of successes out of a known, fixed number of independent attempts suggests binomial; a count of events over a continuous interval of time/space with no natural upper bound suggests Poisson. In practice you rarely just assume — you plot a histogram of the actual data and visually compare it to the candidate distribution's shape, and can formally test the fit with a goodness-of-fit test (a chi-square test is the standard tool, covered in the next submodule).

- 'How many out of a known fixed n ?' → binomial. 'How many in a period, with no fixed upper limit?' → Poisson. 'What's the typical value of a continuous measurement?' → often normal.
- Many real distributions are only approximately normal/binomial/Poisson — the value of naming the closest match is that it lets you borrow well-understood formulas and tests, not that the fit is ever perfect.
- `scipy.stats` provides dozens of other distributions (exponential, uniform, gamma, beta) for cases these three don't fit well.

Common Pitfalls

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- Assuming any bell-shaped-looking histogram is normal without checking — many real distributions are skewed and only superficially resemble a bell curve.
 - Using the binomial distribution when trials aren't actually independent or don't share the same success probability (e.g. conversion rate changes over the campaign).
 - Forgetting that Poisson assumes a constant average rate — if the rate genuinely varies (rush hour vs. midnight), a single λ is insufficient.
 - Confusing a probability density (continuous, normal pdf) with a probability itself — density values from `dist.pdf()` can exceed 1 and are not probabilities on their own; only areas under the curve (via `cdf`) are.
 - Treating the 68-95-99.7 rule as exact for any bell-shaped data — it's exact only for a true normal distribution, and is an approximation elsewhere.

Key Takeaways

- A distribution's family plus its parameters fully describe it — you don't need the raw data once you know both.
- Normal models continuous, symmetric measurements; binomial counts successes out of a fixed number of independent trials; Poisson counts events over an interval with no fixed upper bound.
- The 68-95-99.7 rule and z-scores are the standard shortcuts for reasoning about where a value falls in a normal distribution.
- Poisson's mean and variance are both equal to λ ; λ is the fingerprint that helps confirm it's the right model.
- Always sanity-check a distribution assumption against a histogram of the real data before trusting formulas built on it.

Learn More

SciPy — Statistical Distributions (scipy.stats)

<https://docs.scipy.org/doc/scipy/reference/stats.html>

SciPy — scipy.stats.norm Reference

<https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.norm.html>

Seeing Theory — Probability Distributions

<https://seeing-theory.brown.edu/probability-distributions/index.html>